

Software Design and Analysis QUIZ 01

Name: _____

Program/Sec: _____

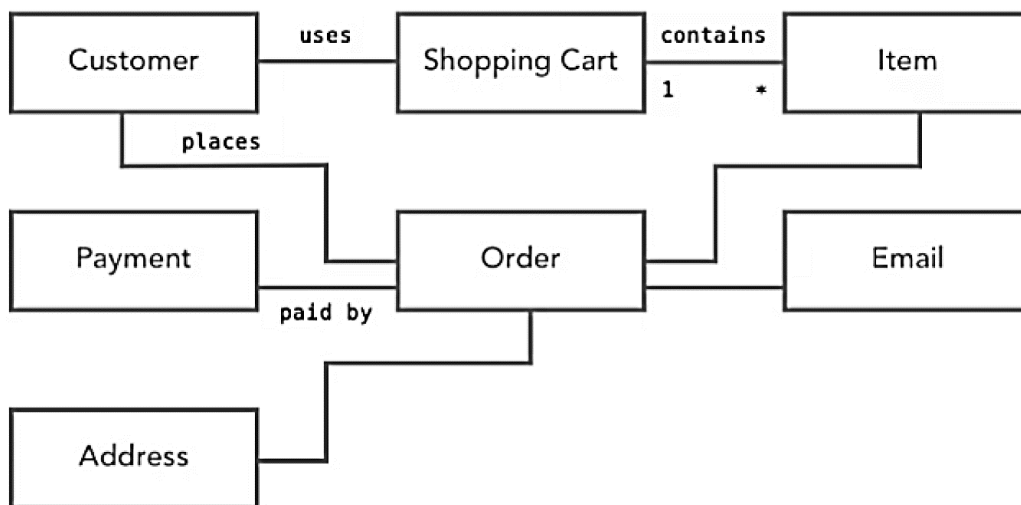
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Scenario

You are a System Modeling & Analysis expert consultant. ABC Company is developing a project based on the following requirement. Initially, an employee was assigned to create a UML diagram for the system, but he resigned midway through the project. The company has now engaged your expert services to review the existing UML diagram and assess its accuracy. You are asked to identify any mistakes or inconsistencies so that the project can smoothly resume and be delivered on time.

Project Requirement

The customer confirms items in the shopping cart. The customer then provides payment and address details to proceed with the sale. The system validates the payment, confirms the order, and generates an order number that the customer can use to track the order status. Additionally, the system sends the customer a copy of the order details via email.



Questions

Q1. Does the employee's diagram is correct? (if not, answer Q2 & Q3)

a. Yes, diagram is correct

b. No, diagram is not correct

Q2. Does the employee create right type of UML diagram or it has error? (if not, answer Q4 & Q5)

a. Right Type of UML and has no errors

b. Not Right type and has errors

c. Right Type of UML and has errors

d. Not Right type, but has no errors

OR

Ans: The given project requirements represents the user shopping experience which primary concern with user operations/process. The given picture represents structural or data modeling of system, while require behavioral representation.

Q3. What is/are the mistake(s) in the employee's diagram?
list down where and what are the mistakes at back side.

Ans: There almost all relationships are incorrect as behavioral diagram cannot be express as structural model of system. Therefore, the diagram has following incorrect/inappropriate relationship.

- Customer – Shopping cart, Customer – Order, it is not feasible to hold same information in two different tables/entity while referring to same process.
- Order – Address, Order – Email, Order – Payment, order uses them as reference attributes from
- customer not as separate entity.
- Order – Shopping, referring to same thing as two different entities.

Q4. What would be the right type of UML diagram for the given problem (**Name only**)?

Ans: Use-case Diagram

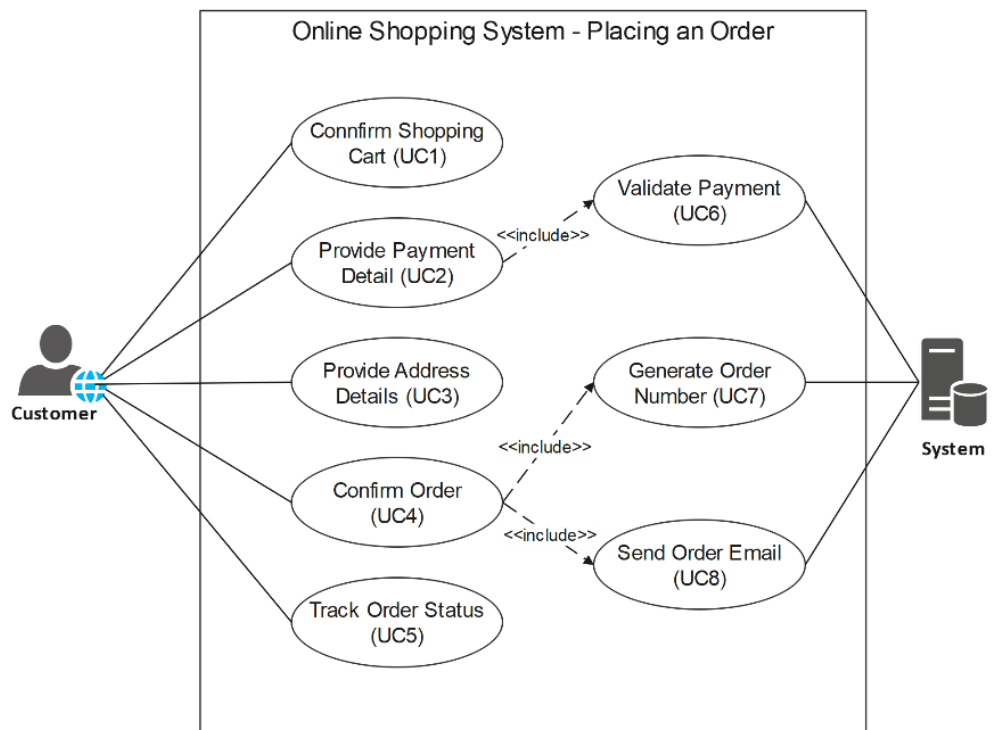
Q5. Draw the right UML diagram.

Ans: UML Use Case Diagram **Actors** are:

- **Customer** — initiates interactions
Confirm items in cart, provide payment details, provide address details, confirm order and track order status.
- **System** — handles automated backend processes
Validate payment, generate order number, and send order details via email.

List of Use Cases

ID	Use Case
UC1	Confirm Shopping Cart
UC2	Provide Payment Details
UC3	Provide Address Details
UC4	Confirm Order
UC5	Track Order Status
UC6	Validate Payment
UC7	Generate Order Number
UC8	Send Order Email



Note: Student answers not restricted to the language and content of the answer except the diagram. If students reflect the same understanding and describe the things correctly will be graded, otherwise not.